Program to demonstrate Simple Inheritance

```
// Create a superclass.
class A {
  int i, j;
  void showij() {
     System.out.println("i and j: " + i + " " + j);
}
}
// Create a subclass by extending class A.
class B extends A {
  int k;
  void showk() {
    System.out.println("k: " + k);
  }
  void sum() {
     System.out.println("i+j+k: " + (i+j+k));
  }
}
public class SimpleInheritance {
public static void main(String[] args) {
     A superOb = new A();
     B subOb = new B();
     // The superclass may be used by itself.
     superOb.i = 10;
     superOb.j = 20;
     System.out.println("Contents of superOb: ");
     superOb.showij();
     System.out.println();
     /* The subclass has access to all public members of
     its superclass. */
     subOb.i = 7;
```

```
    subOb.j = 8;
    subOb.k = 9;
    System.out.println("Contents of subOb: ");
    subOb.showij();
    subOb.showk();
    System.out.println();
    System.out.println("Sum of i, j and k in subOb:");
    subOb.sum();
}
}
```

Program to Illustrate Member Access in Inheritance

```
// Create a superclass.
class A {
  int i; // default access
  private int j; // private to A
  void setij(int x, int y) {
    i = x;
    j = y;
  }
}
// Sub class.
class B extends A {
  int total;
  void sum() {
  total = i + j;
  }
}
public class Access {
public static void main(String[] args) {
  B subOb = new B();
```

```
 subOb.setij(10, 12);
 subOb.sum();
 System.out.println("Total is " + subOb.total);
}
}
```

Program to Demonstrate when constructors are executed.

```
// Create a super class.
class A {
A() {
System.out.println("Inside A's constructor.");
}
}
// Create a subclass by extending class A.
class B extends A {
B() {
System.out.println("Inside B's constructor.");
}
}
// Create another subclass by extending B.
class C extends B {
C() {
System.out.println("Inside C's constructor.");
}
}
class CallingCons {
public static void main(String[] args) {
C c = new C();
}

// Using super to overcome name hiding.
```

```
class A {
int i;
}
// Create a subclass by extending class A.
class B extends A {
int i; // this i hides the i in A
B(int a, int b) {
super.i = a; // i in A
i = b; // i in B
}
void show() {
System.out.println("i in superclass: " + super.i);
System.out.println("i in subclass: " + i);
}
}
class UseSuper {
public static void main(String[] args) {
B subOb = new B(1, 2);
subOb.show();
}
}
```

Program to Illustrate Method overriding.

```
class A {
int i, j;
A(int a, int b) {
i = a;
j = b;
}
// display i and j
void show() {
System.out.println("i and j: " + i + " " + j);
```

```
}
}
class B extends A {
int k;
B(int a, int b, int c) {
super(a, b);
k = c;
}
// display k – this overrides show() in A
void show() {
System.out.println("k: " + k);
}
}
class Override {
public static void main(String[] args) {
B subOb = new B(1, 2, 3);
subOb.show(); // this calls show() in B
}
}
```